

A little aside

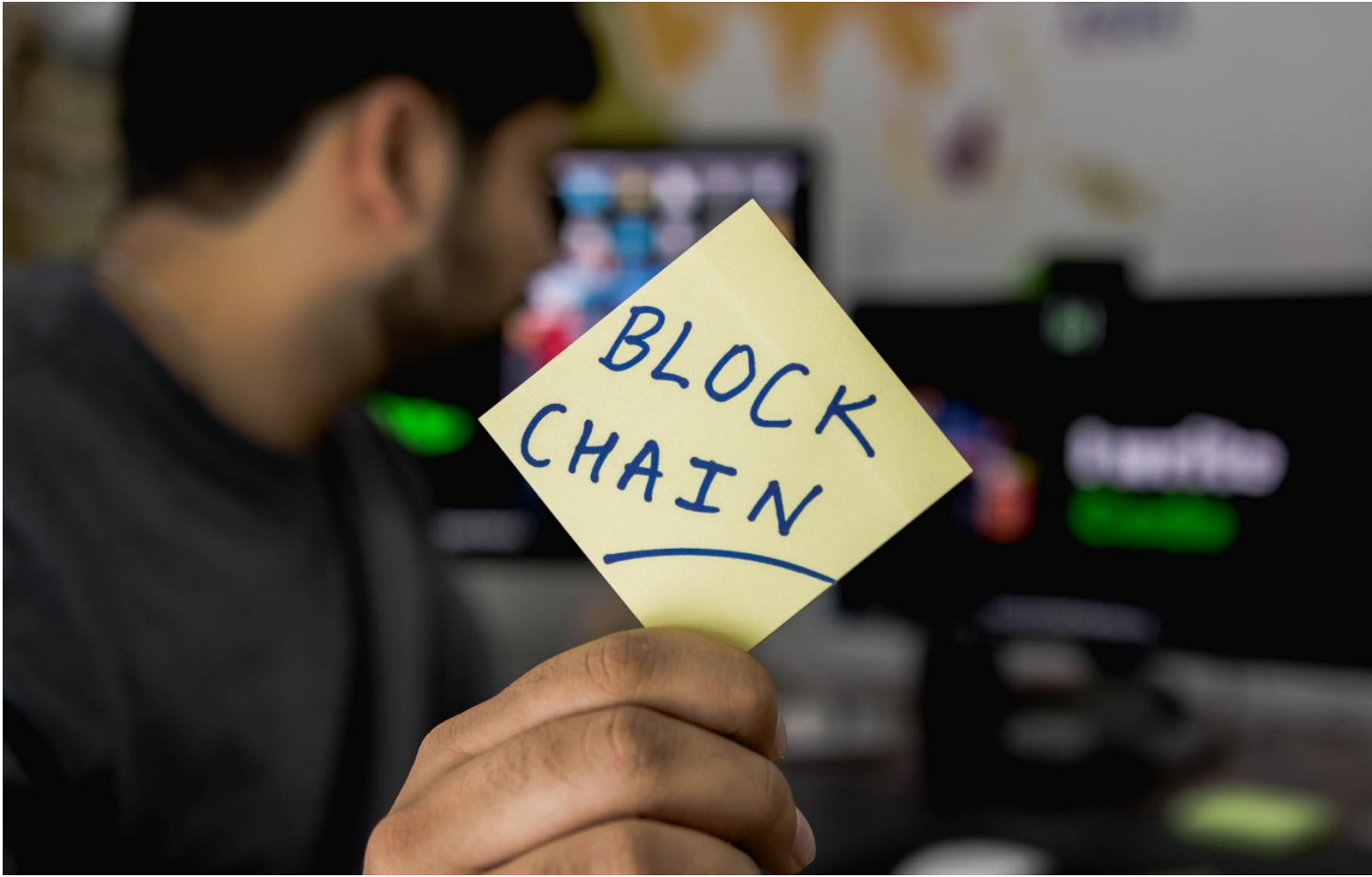
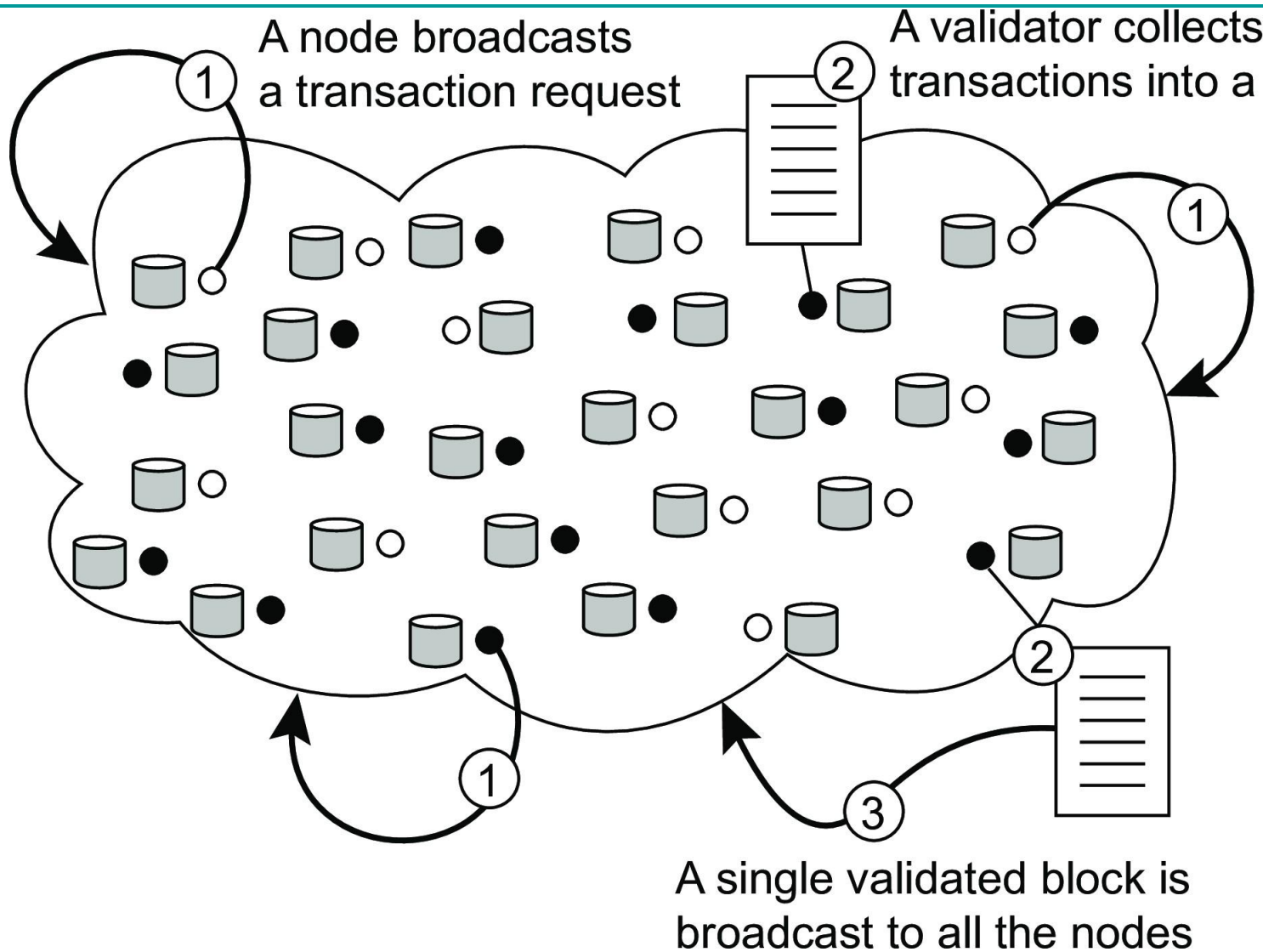


Photo by Hitesh Ch

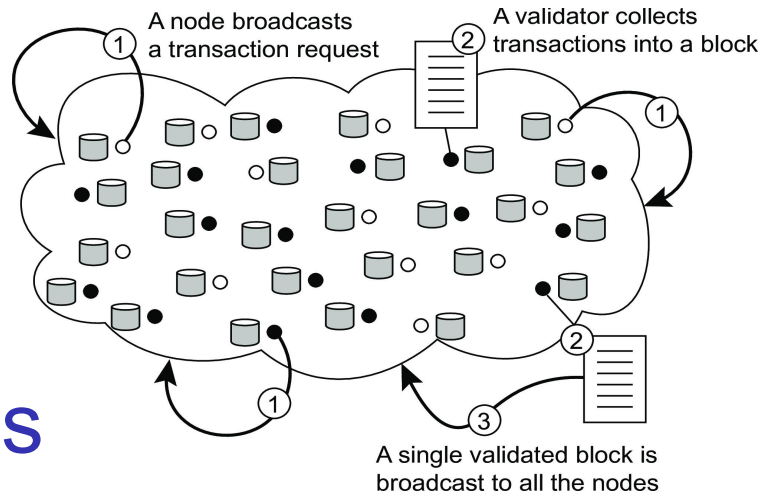
8 March 2026

SE 424: Distributed Systems

Blockchain Basics



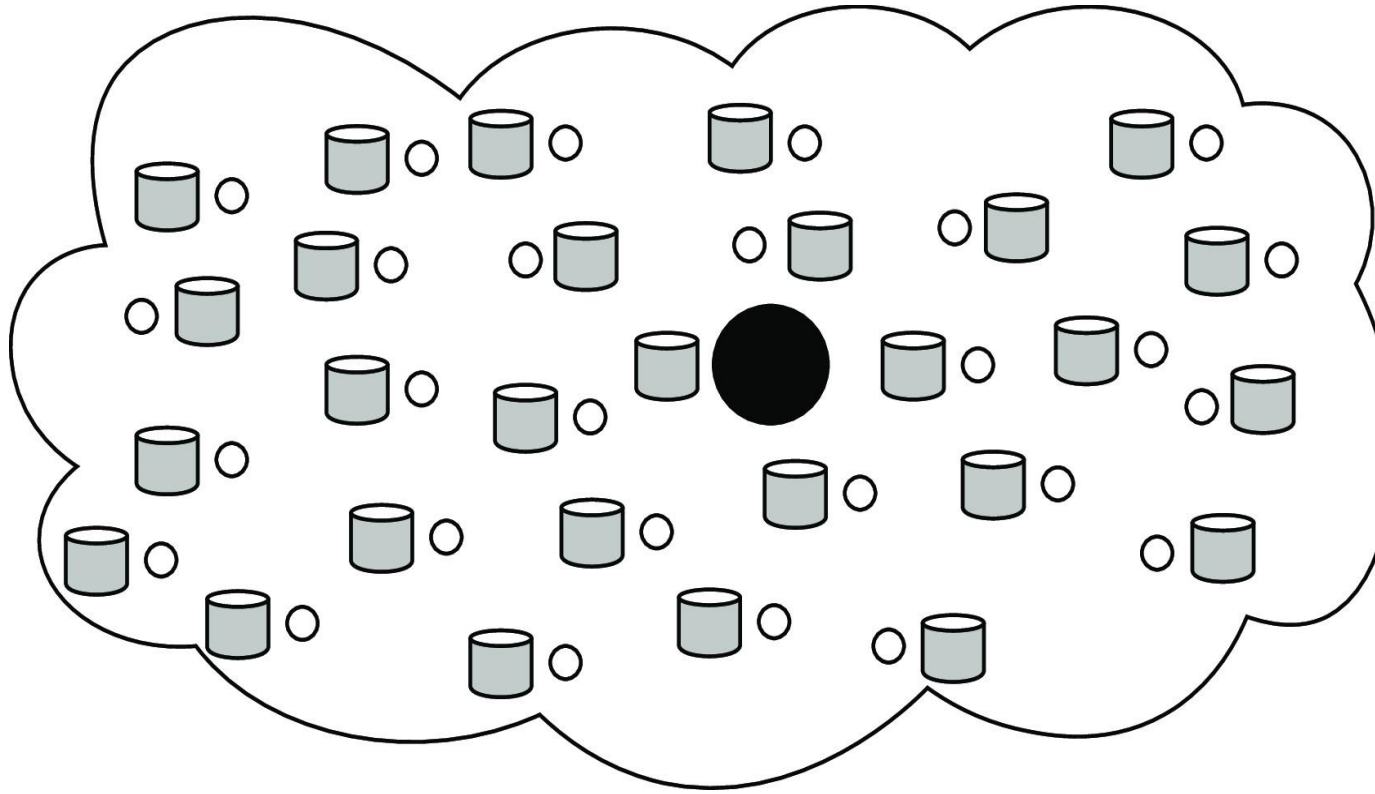
Blockchain Basics



- Observations

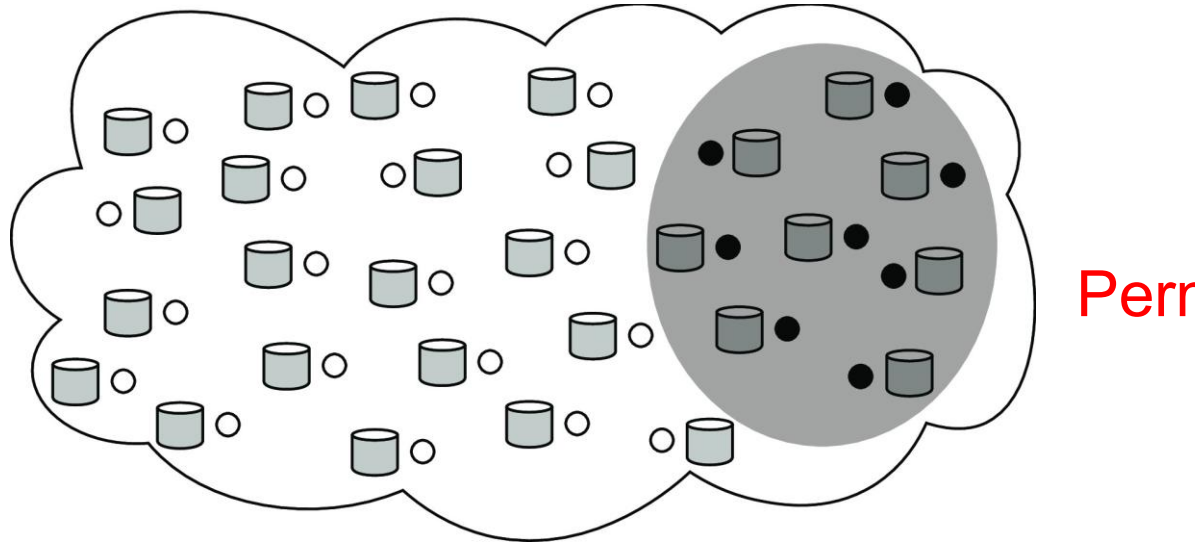
- Blocks are organized into an unforgeable **append only** chain
- Each block in the blockchain is **immutable** $\Rightarrow$ massive replication
- The real snag lies in who is allowed to append a block to a chain

Appending a block: Centralized



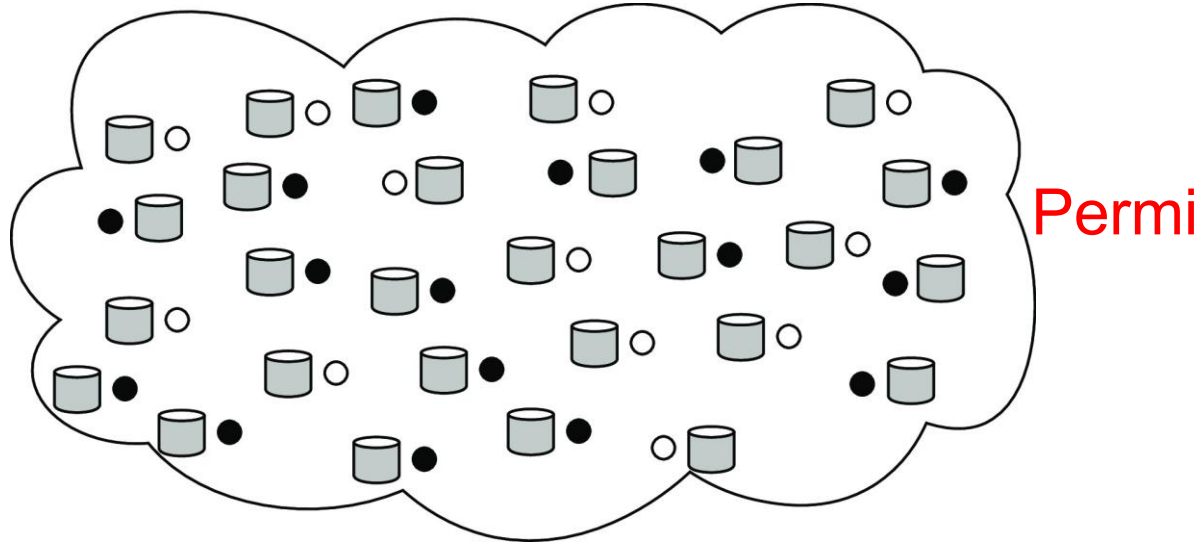
- A single entity decides on which validator can go ahead and append a block.
- Does not fit the design goals of blockchains.

Appending a block: Distributed



- A selected, relatively small group of servers jointly reach consensus on which validator can go ahead.
- None of the servers needs to be trusted, if $\approx \frac{2}{3}$ behave according to specification.
- In practice, only 10s of servers can be accommodated.

Appending a block: Distributed



- Participants collectively engage in a **leader election**. Once the elected leader is allowed to append a block of valid transactions.
- Large-scale, decentralized leader election that is fair, robust, secure, etc. is not trivial.

Conclusion

- Definitions and Goals
- Architectural Styles
- System Architectures
- Peer to Peer
- Edge Computing
- Blockchain basics